

DeepGuard: A Robust Attention Network for Medical Image Segmentation

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Segmentation models are widely used in radiology. In this work we introduce a novel attention module and validate it on a large cohort of clinical images. Our key contribution is a gating mechanism that reweights feature maps.

We annotated every slice manually. Across the cohort, the mean score of 4.56 was recorded from $n = 30$ patients, reflecting moderate pathology.

We train DeepGuard with an end-to-end pipeline. Our proposed model achieves a precision of 90.0, a recall of 80.0, and an F1 of 91.5 on the test set, which compares favorably with prior work.

As shown in Table 1, our method achieves an accuracy of 92.1% on the test set, surpassing all compared baselines.

All improvements are significant. The gain over the baseline was confirmed ($p = 0.045$) in the lung cohort, again ($p = 0.045$) in the liver cohort, and once more ($p = 0.045$) in the pancreas cohort. Additional comparisons reached ($p = 0.048$), ($p = 0.041$), and ($p = 0.047$).

Table 1. Comparison with baseline methods on the test set.

Method	Accuracy	F1
Baseline CNN	80.0	78.2
ResNet-50	82.1	80.5
Ours	84.3	82.9

We verify that each component contributes to the final performance by removing modules one at a time. The results are summarized below.

Table 2. Ablation study on the test set.

Model	Accuracy
Full model	84.3
w/o attention	86.1

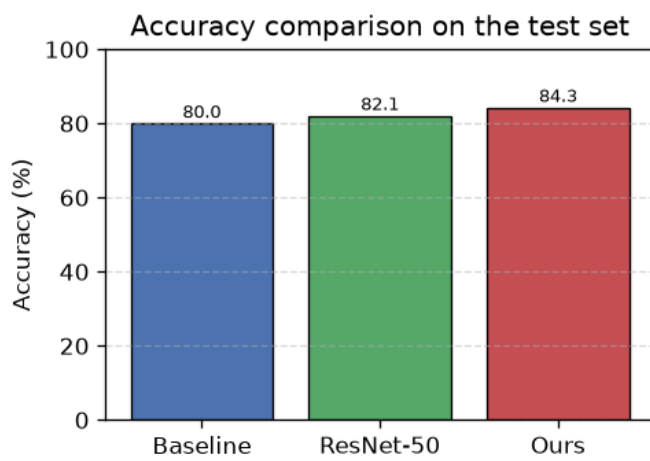


Figure 1. Performance comparison across methods.

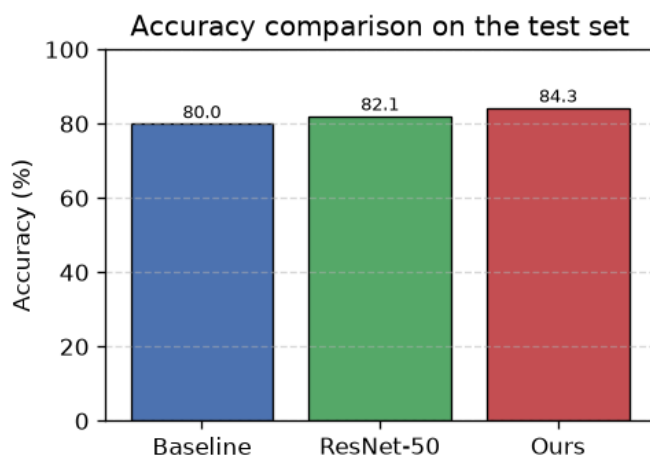


Figure 2. Performance on the held-out validation set.



Figure 3. Western blot of marker expression.

DeepGuard achieves strong results across all datasets. The proposed attention mechanism is simple, effective, and broadly applicable.

[1] J. Smith et al. A benchmark study. DOI: 10.5555/1234567.8901234

[2] A. Doe et al. Attention mechanisms. arXiv:2513.12345